

5. Introduction of Programming and Software

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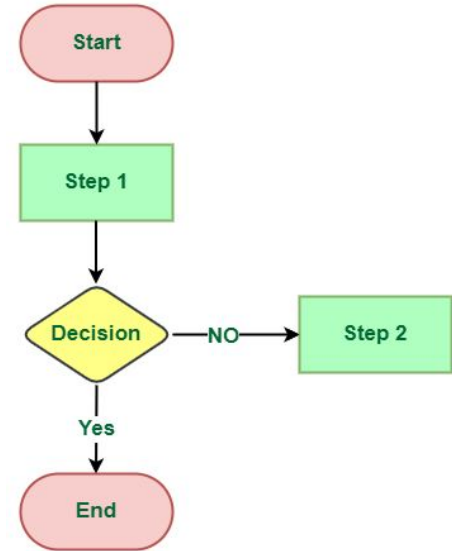
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- 1. Algorithm**
- 2. Programming Language**
- 3. Code**
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- 5. Software**

1. Algorithm

- An algorithm is a step-by-step set of instructions to solve a problem or complete a task. It is a plan.
- An algorithm is the idea or the logic behind the solution. It's not written in a programming language yet; it can be in **plain English or a flowchart**. It's the "what" and "how" of the process.
- **Key Characteristics:** It's unambiguous, well-defined, and has a clear starting and ending point.



Algorithm, Pseudocode & Flowchart - 1

Algorithm & Flowchart to find the sum of two numbers

Algorithm

Step-1 Start

Step-2 Input first numbers say A

Step-3 Input second number say B

Step-4 $SUM = A + B$

Step-5 Display SUM

Step-6 Stop

Pseudocode

Begin

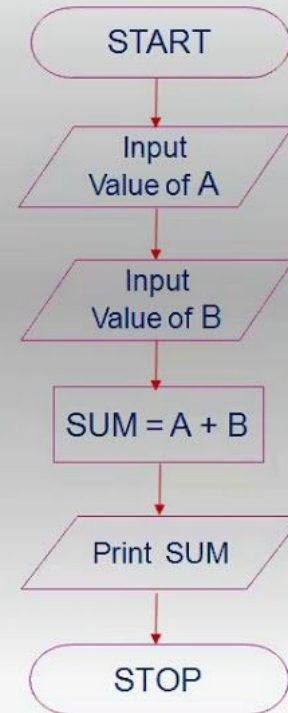
Input A

Input B

Compute $SUM = A + B$

Print SUM

End



2. Programming Language

- A **programming language** is a formal language used to write programs.
- It is used to write instructions that a computer can understand and execute.
- It provides the syntax (grammar) and keywords (vocabulary) that a programmer uses to write code. Each language has its own strengths and is suited for different tasks (e.g., web development, data science, mobile apps).

2. Programming Language

Front-end web development



JavaScript



Elm



TypeScript



Cascading
Style Sheets (CSS)

Back-end web development



JavaScript



TypeScript



Python



Go



Ruby



Scala

Mobile development



Swift



Java



C#

Game development



C#

Desktop applications



Java



Python



JavaScript



TypeScript



Scala



Go

Systems programming



C#



C++



Go

3. Code

- Code (or source code) is the set of instructions written in a specific programming language that implements an algorithm.
- Code is the text file that a programmer creates. It is human-readable (for those who know the language) but cannot be directly executed by the computer's brain (the CPU).

4. Program

- A program is the final, runnable product after the code has been compiled or is being executed by an interpreter (**compiler/interpreter = translator**). It is a sequence of machine code instructions that the computer's CPU can understand and execute directly.
- When we double-click on `chrome.exe` or `notepad.exe`, we are running a program. So, a Program is the executable file we run to perform a specific task.

```
#include <stdlib.h>
#include <stdio.h>

int main (void) {
    printf("Hello, World!\n");
    exit(0);
}
```

Fig: Code in C language

5. Software

- Software is a collection of one or more programs, along with associated data and documentation, designed to fulfill a particular purpose for the user.
- So, Software is the entire suite of programs and resources that provide a full-featured application.
- For example, Microsoft Word is software. It consists of the main program (winword.exe), helper programs for spell check, printing, etc., and all its data files (templates, dictionaries). A video game, our mobile banking app, and the Windows operating system itself are all examples of software.

5. Software

Let's tie it all together with an analogy of building a house i.e. **Software**:

1. **Algorithm**: The architect's plan and blueprint for the house. (*The idea*)
2. **Programming Language**: The set of construction rules and terminology the builders will use (e.g., metric system, specific building codes). (*The rules of communication*)
3. **Code**: The detailed work orders and instructions written for the construction crew, based on the blueprint and using the correct terminology. (*The written instructions*)
4. **Compiler/Interpreter**: The general contractor who interprets the work orders and directs the crew to execute them. (*The translator/manager*)
5. **Program**: A single, finished component of the house, like the fully constructed roof or the installed plumbing system. (*A functional component*)
6. **Software**: The entire, finished house, complete with roof, plumbing, electrical systems, paint, and landscaping, ready for you to live in. (*The final product*)

END